

MODULE 3A • EPISODE 2

Making Measurements

Significant Figures, Precision, and Units

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Significant Figures, Precision, and Units

This episode picks up where the last one left off and asks how to write a measurement honestly. The panel starts with significant figures — the digits you know for sure plus one guessed digit — and shows why scientific notation clears up the confusion caused by leading and trailing zeros. The discussion then turns to the difference between precision and accuracy, and to systematic errors: trivial ones like zero error and parallax, which should simply be corrected, and non-trivial ones built into the equipment or procedure. The second half covers units: the US-English and metric systems, the MKS and CGS varieties of the metric system, and the conversion factors that link them. The episode closes with the rules for carrying significant figures through calculations, so that a calculator's ten digits never claim more precision than the measurements justify.

The Panel

Name	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Guest; cosmologist, European Space Agency
Dr. Amara Johnson	Guest; theoretical physicist

Learning Objectives

After this episode you should be able to:

- Define significant figures and identify the least significant digit of a measurement.
- Write a measurement in scientific notation and explain why it removes ambiguity.
- Count significant figures, including leading and trailing zeros.
- Distinguish precision from accuracy, and explain why precision is necessary but not sufficient for accuracy.
- Tell trivial systematic errors (zero error, parallax) from non-trivial ones, with an example of each.
- Explain parallax and use proper technique to minimize it when reading analog devices.
- Name the US-English and MKS units for length, time, mass, energy, force, and charge.
- Use conversion factors within and between unit systems.
- Round the result of a calculation to the correct number of significant figures.

QUICK REFERENCE

Key Terms

Term	Definition
Significant figures (sig figs)	All the digits in a measurement that you know for sure, plus one more digit that is guessed or uncertain.
Least significant digit	The rightmost significant digit; the one that was guessed, or uncertain.
Scientific notation	A number between 1 and 10, not including 10, multiplied by the necessary power of ten.
Standard notation	A number written out in ordinary form, such as 0.0069 or 75,000,000.
Leading zeros	Zeros to the left of the first nonzero digit. In standard notation they are never significant; they are placeholders.
Trailing zeros	Zeros at the right end of a number. In standard notation they may or may not be significant; in scientific notation they always are.
Accuracy	How close a measurement is to the true value.
Precision	How finely you can measure (how many significant figures you can legitimately write) and how closely repeated measurements agree.
Systematic error	An error that pushes measurements consistently in one direction.
Trivial systematic error	Only nominally a systematic error; really a mistake or blunder that should be corrected, such as zero error or parallax.
Zero error	An error introduced when a device does not read zero when it should.
Non-trivial systematic error	An inherent limitation in the equipment or procedure of an experiment; it can sometimes be accounted for, but not really corrected.
Reaction-time and anticipation errors	Errors built into hand timing: starting or stopping a timer a little late, or a little early if you anticipate the signal.
Parallax	The apparent shift of a nearby object against a more distant background when the viewing position, or the eye used, changes.
Physical unit	A standardized amount of a physical quantity.
US-English system	The system of units (feet, pounds, seconds) used almost exclusively in the United States; sometimes mistakenly called the Imperial system.
MKS system	The metric system based on meters, kilograms, and seconds; the basis of the SI. Used in physics.
CGS system	The metric system based on centimeters, grams, and seconds. Common in chemistry.
SI	The International System of Units, from the French <i>Système International</i> .
Slug / pound-mass	Two competing units of mass used in the US-English system.
Conversion factor	A ratio equal to the number one, used to change one unit to another.
Linking conversion factor	A conversion factor that connects the US-English and metric systems, such as 2.54 cm per 1 inch.
Linear mass density	Mass per unit length, $D = m/L$.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
4	Significant digits in 12.34 cm; the 4 is the least significant digit
2	Significant digits in 0.0069 cm (the 6 and the 9)
75,000,000 yr	Dinosaur-bone age in standard notation; the trailing zeros are ambiguous
7.50×10^7 yr	The same age in scientific notation: three significant figures, final zero uncertain
12.34 ± 0.01 cm	First measurement in the precision example
12.345 ± 0.001 cm	More precise measurement, but not necessarily more accurate
12 in = 1 ft	Also 3 ft = 1 yd and 5,280 ft = 1 mi
1000	mm per m, m per km, and g per kg in the metric system
2.54 cm = 1 in	Linking conversion factor
39.37 in = 1 m	Linking conversion factor
1.609 km = 1 mi	Linking conversion factor
2.205 lb = 1 kg	Convenient, but not “kosher”: it turns a unit of force into a unit of mass
60 / 60 / 24	Seconds per minute, minutes per hour, hours per day
365.25 days = 1 yr	The Julian calendar approximation
12.3 g, 2.964 cm	Mass (3 sig figs) and length (4 sig figs) of the thin rod
4.149797571 g/cm	Calculator result for $D = m/L$: ten digits, most of them meaningless
4.15 g/cm	The correctly reported linear mass density

EPISODE CONTENT

Part-by-Part Summary

Part 1 — Significant Figures

Significant figures, also called significant digits or “sig figs,” are all the digits you know for sure, plus one more digit that you guess or that is in some way uncertain. If your best guess for a measurement is 12.34 cm, there are four significant digits. The rightmost digit, the 4, is the **least significant digit**: the one that was guessed.

Significant figures tell us about the **precision** of a measurement — how many meaningful digits we can legitimately claim to know. Using them properly takes practice and patience.

The first and most important rule: express measurements in **scientific notation**, as a number between 1 and 10 (not including 10) multiplied by the necessary power of ten. **Standard notation** is often confusing. **Leading zeros** in standard notation are never significant: 0.0069 cm has only two significant digits, the 6 and the 9. The zeros are just placeholders.

Part 2 — Trailing Zeros and Scientific Notation

Trailing zeros in standard notation may or may not be significant. If a dinosaur bone is reported to be 75,000,000 years old, how many of those zeros were actually measured, and how many are just placeholders? There is no way to tell.

Trailing zeros in scientific notation are always significant, because otherwise you simply wouldn't write them. Writing the bone's age as 7.50×10^7 years states explicitly that there are three significant figures and that the zero after the 5 is the guessed, least significant digit. Scientific notation removes the ambiguity.

Part 3 — Precision and Accuracy

Accuracy is how close a measurement is to the true value. **Precision** is how finely you can measure — how many significant figures you can legitimately write — and how closely repeated measurements agree with one another. The two are often confused.

It is tempting to think that 12.345 ± 0.001 cm must be closer to the true value than 12.34 ± 0.01 cm because its confidence interval is smaller. Not necessarily: a very precise measurement can be quite inaccurate if the device was not properly calibrated — for example, if it has an uncorrected **zero error**.

The correct relationship: being precise is **necessary, but not sufficient**, for being accurate. A device that is systematically off by a fixed amount gives precise measurements that are all consistently wrong — high precision, low accuracy. A calibrated but imprecise device gives measurements that scatter around the true value — better accuracy on average, but lower precision.

Part 4 — Systematic Errors and Parallax

There are two main categories of **systematic errors**. A **trivial systematic error**, such as a zero error, is only nominally a systematic error; it is really a mistake, a blunder, and should be corrected as a matter of course.

A **non-trivial systematic error** is an inherent limitation in the equipment or procedure. It can't really be corrected, though it can sometimes be accounted for in the analysis. Starting and stopping a timer by hand builds **reaction-time and anticipation errors** into every time measurement: you may consistently be a little late responding to a voice command, or a little early if you anticipate it.

Parallax, or eyeball error, is another trivial systematic error. Nearby objects appear to shift against more distant ones depending on where you view them from, including which eye you use: hold your thumb at arm's length, close each eye in turn, and your thumb jumps against the wall. Parallax only affects **analog** devices read by eye, where an edge or needle is compared with a ruled scale.

To minimize it, line your nose up directly over the measurement and alternate eyes; the reading should shift the same amount to each side. With “one good eye,” use that eye directly over the measurement, looking straight down at the scale. Above all, develop a consistent technique.

Part 5 — Units and Systems of Units

A **physical unit** is a standardized amount of a physical quantity. Units let people agree on how much of something was bought at a grocery store, and the same principle applies to experiments: without standardized units, scientific communication would be impossible.

Two main systems are in use today. The **US-English system**, sometimes mistakenly called the Imperial system, is used almost exclusively in the United States, with some lingering use of the Imperial system in the United Kingdom. The **metric system** comes in two varieties: **MKS** (meters, kilograms, seconds), the basis of the **SI**, used almost exclusively in physics; and **CGS** (centimeters, grams, seconds), whose grams and cubic centimeters are common in chemistry.

The most frequently measured quantities are length (feet vs. meters), time (seconds in both), mass (kilograms in MKS), energy (foot-pounds vs. joules), force (pounds vs. newtons), and charge (coulombs in both). The US-English system was designed around force rather than mass, so the pound is technically a unit of force. Some textbooks use the **slug** for mass, others the **pound-mass** — one more reason the metric system is preferred in science.

Part 6 — Conversion Factors and Calculations

A **conversion factor** is a ratio equal to the number one that changes one unit to another: 12 inches per 1 foot, 3 feet per 1 yard, 5,280 feet per 1 mile; 1000 millimeters per 1 meter, 1000 meters per 1 kilometer, 1000 grams per 1 kilogram. The metric system is simpler because it is all based on powers of 10.

Linking conversion factors connect the two systems: 2.54 cm per 1 inch, 39.37 inches per 1 meter, 1.609 km per 1 mile. The factor 2.205 pounds per 1 kilogram is not considered “kosher,” because it changes a unit of force into a unit of mass, but it is very convenient, so we use it. Time factors are shared by both systems: 60 s per minute, 60 min per hour, 24 h per day, and 365.25 days per year, the Julian calendar approximation.

A thin rod has a mass of 12.3 g (three significant figures) and a length of 2.964 cm (four). Its **linear mass density** $D = m/L$ comes out on a calculator as 4.149797571 g/cm. But a calculator can't create

significant digits out of thin air: the uncertainty in each measurement's least significant digit “taints” all but the leftmost digits of the result, so the answer is $D = 4.15 \text{ g/cm}$.

For **multiplication and division**, the result can have no more significant figures than the measurement with the fewest. For **addition and subtraction**, the result has the same number of decimal places as the measurement with the fewest decimal places. These rules keep us honest about the limitations of our data.

The next episode ventures into astronomical measurements, where the scales and techniques are quite different from measurements here on Earth.

STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Significant Figures

- 1.** What are significant figures (also called significant digits or “sig figs”)?

ANSWER All the digits you know for sure, plus one more digit that you “guess” or that is in some way uncertain.

- 2.** A measurement is recorded as 12.34 centimeters. How many significant digits does it have, and which digit is the least significant digit?

ANSWER Four significant digits. The rightmost digit, the 4, is the least significant digit; it is the one that was guessed, or uncertain.

- 3.** What do the significant figures of a measurement tell us?

ANSWER They tell us about the precision of the measurement: how many meaningful digits we can legitimately claim to know.

- 4.** What is scientific notation, and why should measurements be expressed in it?

ANSWER Scientific notation writes a number as a value between 1 and 10 (not including 10) multiplied by the necessary power of ten. It eliminates much of the confusion about significant figures that arises in standard notation.

- 5.** How many significant digits are in 0.0069 centimeters? Explain.

ANSWER Two: the 6 and the 9. Leading zeros in standard notation are never significant; they are just placeholders.

Part 2 — Trailing Zeros and Scientific Notation

- 1.** Why are trailing zeros in standard notation confusing? Use the example of a dinosaur bone reported to be 75,000,000 years old.

ANSWER Trailing zeros in standard notation may or may not be significant. For 75,000,000 years, you cannot tell how many of the zeros were actually measured and how many are just placeholders.

- 2.** Are trailing zeros in scientific notation significant? Why?

ANSWER Yes, always, because otherwise you simply wouldn't write them.

- 3.** If the bone's age is written as 7.50×10^7 years, how many significant figures does it have, and which digit is the uncertain one?

ANSWER Three significant figures. The trailing zero after the 5 is explicitly the guessed, or least significant, digit.

Part 3 — Precision and Accuracy

1. Define accuracy.

ANSWER Accuracy is how close your measurement is to the true value.

2. Define precision. What two ideas does it include?

ANSWER Precision is about how finely you can measure (how many significant figures you can legitimately write) and how closely repeated measurements agree with one another.

3. A length is first recorded as 12.34 ± 0.01 cm, and later, with a more precise device, as 12.345 ± 0.001 cm. Must the second measurement be closer to the true value? Explain.

ANSWER No. The second is more precise (it has a smaller confidence interval), but a very precise measurement can be quite inaccurate if the device was not properly calibrated, for example if it has a zero error that hasn't been corrected or accounted for.

4. What is the correct way to describe the relationship between precision and accuracy?

ANSWER Being precise is necessary, but not sufficient, for being accurate. You need precision to achieve accuracy, but precision alone doesn't guarantee accuracy.

5. Describe a situation with high precision but low accuracy, and one with better accuracy but lower precision.

ANSWER High precision, low accuracy: a measuring device systematically off by a fixed amount gives very precise measurements that are all consistently wrong by that same amount. Better accuracy, lower precision: a properly calibrated but not very precise device gives measurements that scatter around the true value.

Part 4 — Systematic Errors and Parallax

1. What are the two main categories of systematic errors?

ANSWER Trivial and non-trivial systematic errors.

2. What is a trivial systematic error? Give an example and explain what should be done about it.

ANSWER It is only nominally a systematic error; it is really a mistake, a blunder. A zero error is an example (parallax is another). It should be corrected as a matter of course.

3. What is a non-trivial systematic error? Describe the example given in the episode.

ANSWER An inherent, built-in limitation in the equipment or the procedure of an experiment. It can't really be corrected, though it can sometimes be accounted for in the analysis. Example: starting and stopping a timer by hand builds reaction-time and anticipation errors into the measurements; you may consistently be a little late responding to a voice command, or a little early if you anticipate it.

4. What is parallax? Describe the thumb demonstration.

ANSWER Parallax is a visual phenomenon whereby nearby objects appear to shift position relative to more distant objects, depending on where you view them from, including which eye you use. Hold your thumb at arm's length in front of your nose and look past it at the wall; close one eye, then open it and close the other, and your thumb appears to jump to one side or the other against the wall.

5. When does parallax error occur in measurement, and what factors affect it?

ANSWER It occurs only when you use your eyes to read analog measuring devices, comparing an edge or a needle with a ruled scale. The reading can shift erroneously to one side depending on how you line your eyes up with the measurement, how good your eyes are, and whether one eye has better vision than the other.

6. How can parallax errors be minimized?

ANSWER With equally good vision in both eyes, line your nose up directly over the measurement and open and close your eyes alternately; the reading should appear to shift the same amount to each side. With “one good eye,” use just that eye, directly over the measurement, looking straight down at the scale rather than at an angle. Above all, be aware of the error and develop a consistent technique.

Part 5 — Units and Systems of Units

1. What is a physical unit, and why are units necessary?

ANSWER A standardized amount of a physical quantity. Units let people agree on how much of something was involved, as in buying meat or vegetables at a grocery store, and the same principle applies in science; without standardized units, scientific communication would be impossible.

2. What are the two main systems of units in use today, and where is each used?

ANSWER The US-English system (sometimes mistakenly called the Imperial system), used almost exclusively in the United States, with some limited, lingering use of the Imperial system in the United Kingdom; and the metric system, which comes in two varieties.

3. What are the two varieties of the metric system, and which sciences tend to use each?

ANSWER MKS (meters, kilograms, seconds), the basis of the SI (Système International), used almost exclusively in physics; and CGS (centimeters, grams, seconds), whose units such as grams and cubic centimeters are very commonly used in chemistry.

4. Give the US-English and MKS units for length, time, mass, energy, force, and charge.

ANSWER Length: feet vs. meters. Time: seconds in both. Mass: kilograms (MKS; no unit given for US-English). Energy: foot-pounds vs. joules. Force: pounds vs. newtons. Charge: coulombs in both.

5. Why was no unit of mass given for the US-English system?

ANSWER The system was originally designed around force rather than mass. The pound is technically a unit of force, though people often treat it as mass; some textbooks use the slug for mass and others the “pound-mass.” This disagreement is one more reason the metric system is preferred in scientific work.

Part 6 — Conversion Factors and Calculations

1. What is a conversion factor? Give examples within the US-English system and within the MKS metric system.

ANSWER A ratio equal to the number one, used to change one unit to another. US-English: 12 inches per 1 foot, 3 feet per 1 yard, 5,280 feet per 1 mile. MKS: 1000 millimeters per 1 meter, 1000 meters per 1 kilometer, 1000 grams per 1 kilogram. The metric system is simpler because it is all based on powers of 10.

2. Name three linking conversion factors between the two systems. Why is 2.205 pounds per 1 kilogram not considered “kosher”?

ANSWER 2.54 centimeters per 1 inch; 39.37 inches per 1 meter; 1.609 kilometers per 1 mile. The factor 2.205 pounds per 1 kilogram changes a unit of force in the US-English system into a unit of mass in the MKS system, but it is very convenient, so we use it.

3. List the time conversion factors given in the episode. What does 365.25 days per 1 year represent?

ANSWER 60 seconds per 1 minute, 60 minutes per 1 hour, 24 hours per 1 day, and 365.25 days per 1 year. The last is the Julian calendar approximation. These factors are shared by both systems.

4. A thin rod has a mass of 12.3 grams and a length of 2.964 centimeters. A calculator gives $D = m/L = 4.149797571$ grams per centimeter. What is the correct answer, and why?

ANSWER $D = 4.15$ g/cm. The mass has three significant figures and the length has four, so the answer can have only three, the lesser of the two. The uncertainty in each measurement's least significant digit “taints” all but the leftmost digits of the result; the answer may contain only digits known for sure plus one guessed (“tainted”) digit.

5. State the significant-figure rules for (a) multiplication and division and (b) addition and subtraction.

ANSWER (a) The result can have no more significant figures than the measurement with the fewest significant figures. (b) The result should have the same number of decimal places as the measurement with the fewest decimal places.

6. Why do these significant-figure rules matter?

ANSWER They ensure that we don't claim more precision in calculated results than our original measurements actually justify; they are a way of being honest about the limitations of our data.

TEACHER NOTES

Corrections and Cautions

Points in the episode that need a correction, a qualification, or a prepared answer for the question a sharp student will ask.

C1 Is the pound really a unit of force?

Since 1959 the avoirdupois pound has been legally defined as exactly 0.45359237 kg — a unit of mass. The unit of force is the pound-force (lbf). The episode's view reflects the older engineering convention. Read with that in mind, 2.205 lb per kg is a legitimate mass-to-mass conversion; it is “not kosher” only if the pound is taken to be a force.

C2 Exact conversion factors don't limit significant figures.

Defined factors such as 12 in per 1 ft, 1000 g per 1 kg, 60 s per 1 min, and 2.54 cm per 1 in (exact by definition since 1959) have unlimited significant figures. Factors like 39.37, 1.609, and 2.205 are rounded, and do count. Students who apply the multiplication rule to a defined factor will throw away good digits.

C3 Two meanings of “precision.”

The episode defines precision as both resolution (how many digits you can write) and repeatability (how closely repeated measurements agree). Many lab courses use only the second meaning. Worth naming both so students aren't caught off guard later.

C4 Necessary but not sufficient — for what?

The episode says precision is necessary for accuracy, yet also describes an imprecise, calibrated device as having “better accuracy on average.” The two fit together once you note that the second statement is about the average of many readings, while the first is about a single measurement.

C5 7.50×10^7 is a choice, not a translation.

Nothing in “75,000,000” tells us there are three significant figures. The episode is showing what someone would write if they knew the zero after the 5; 7.5×10^7 or 7.500×10^7 would be equally valid if that were the actual precision.

C6 Parallax can also be random.

Parallax is systematic when the eye is consistently off to one side. If the viewing angle varies from reading to reading, it shows up as random scatter instead. Either way, good technique removes it.

C7 Hand timing errors are partly random.

Reaction time has a consistent average delay (systematic) plus reading-to-reading variation (random). The episode treats only the systematic part.

C8 The “taint” argument is a rule of thumb.

Significant-figure rules approximate formal uncertainty propagation. For the rod, 12.3 ± 0.1 g gives a relative uncertainty near 0.8%, which supports three significant figures in 4.15 g/cm — the rule and the formal method agree here, but not always.

C9 Years and calendars.

The Julian year is 365.25 days; the Gregorian calendar averages 365.2425 days, and the tropical (seasonal) year is about 365.2422 days. 365.25 is the standard value for converting years in astronomy.

TEACHER NOTES

Teaching the Episode

Start with a calculator display.

Put 4.149797571 g/cm on the board and ask how many of those digits the class believes. Working back to 4.15 g/cm from the 12.3 g and 2.964 cm measurements makes the multiplication rule stick.

Count zeros together.

Give a short list — 0.0069, 75,000,000, 7.50×10^7 , 1200, 1.200×10^3 — and have students count significant figures, then rewrite each ambiguous one in scientific notation.

Run the thumb demonstration.

Have everyone do the thumb-and-wall parallax demonstration, then read a meter stick or graduated cylinder from above, straight on, and below. Seeing the reading shift makes the technique memorable.

Draw the targets.

Sketch four dartboards: precise and accurate, precise but not accurate, accurate on average but not precise, and neither. Ask which one a miscalibrated caliper produces.

Practice conversions with units attached.

Have students convert a height in feet and inches to centimeters and a distance in miles to kilometers, writing every conversion factor as a fraction equal to one and canceling units.

Set up the next episode.

The next episode turns to astronomical measurements. Ask what units would make sense for the distance to the Moon or to a nearby star, and why scientific notation becomes essential.